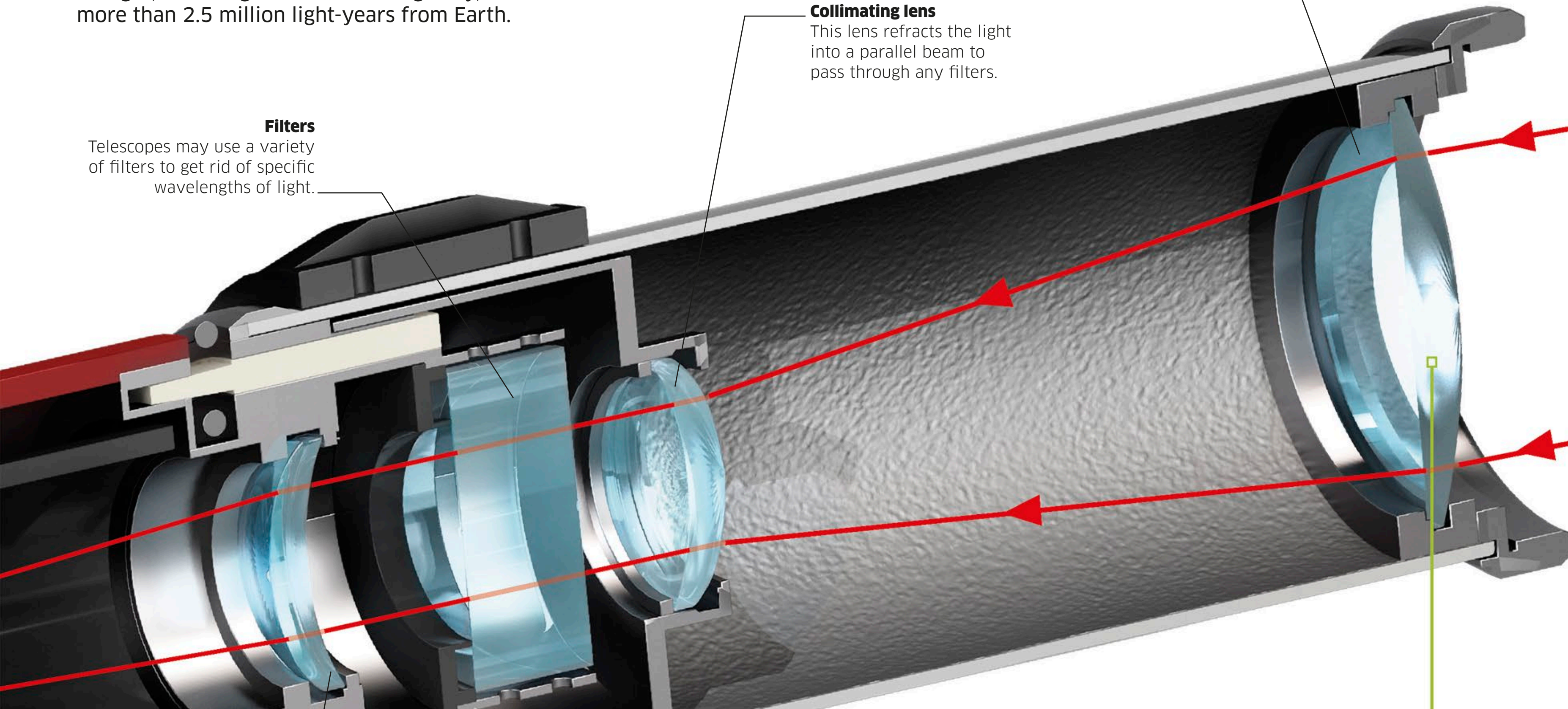
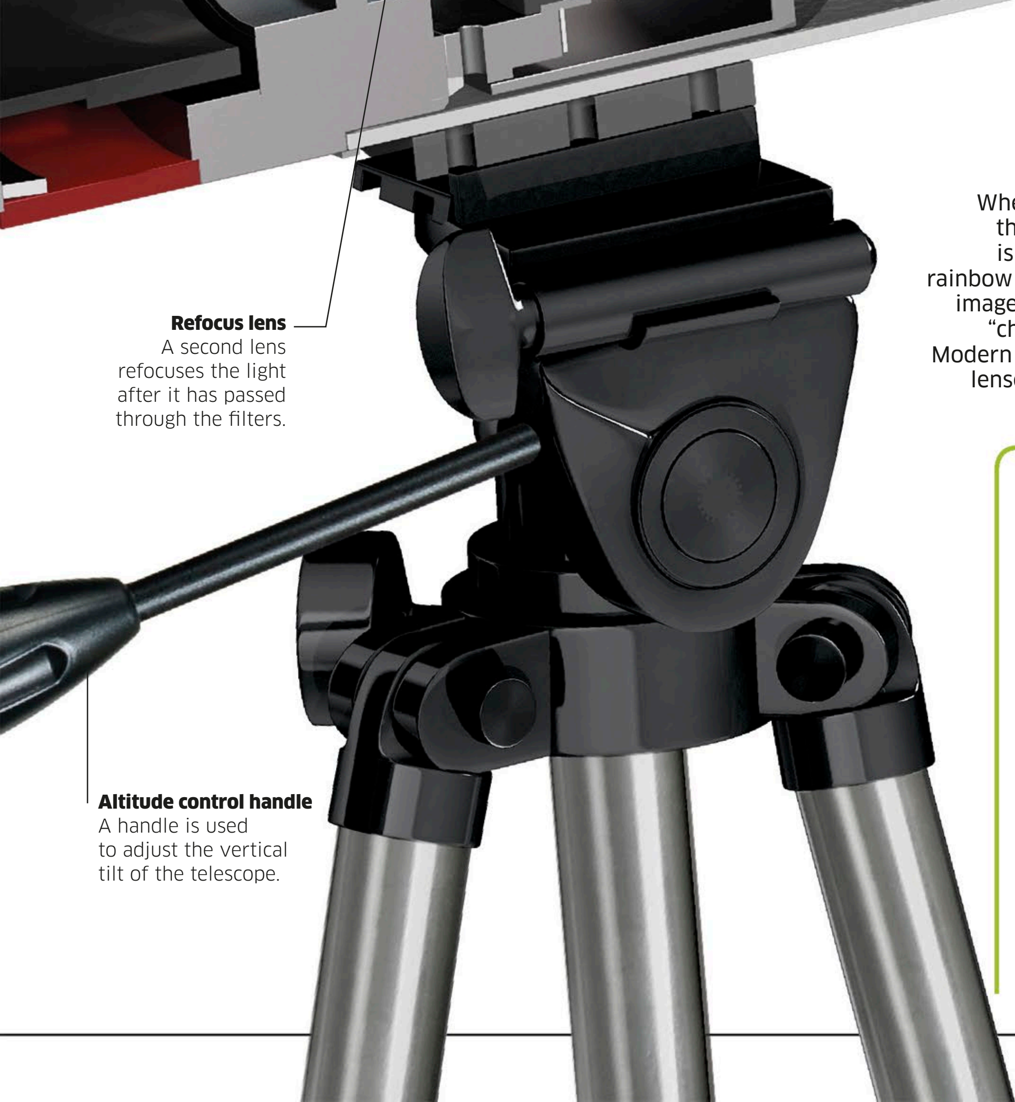
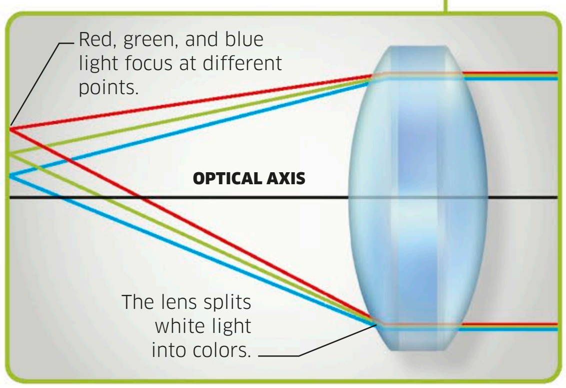
Refracting telescope

This type of telescope uses a convex lens to gather and focus as much light as possible from the distant object. It can be used to look at anything bright enough for light to reach us at night, including the Andromeda galaxy, more than 2.5 million light-years from Earth.



Rainbow effect
When white light passes through a glass lens, it is refracted, creating a rainbow of colors around the image—an effect known as "chromatic aberration." Modern telescopes use extra lenses to counteract this.

Concave and convex mirrors

An image reflected in a concave mirror appears small and, depending how far the viewer is from the mirror, may be upside down. The image in a convex mirror is formed by a virtual image behind the mirror, and appears large.